EXAM PAPERS PRACTICE

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Practice questions created by actual examiners and assessment experts

Detailed mark scheme

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AQA A Level Further Mathematics (7367)

Topic C: Matrices

Sub topic C3: Matrix Transformations (2D)

Mark Scheme

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Mark Scheme

Topic	**C: Matrices**
Sub topic	**C3: Matrix Transformations (2D)**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

EXAM PAPERS PRACTICE

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Obtains two equations in x and y. May be seen as a single vector equation. At least one equation must be correct. Accept a pair of letters other than x and y. Ignore any subsequent incorrect working.	AO1.1a	M1	$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$ $\begin{bmatrix} 2x+3y \\ x+4y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$ $2x + 3y = x$ and $x + 4y = y$
Obtains any two correct invariant points, with no incorrect points. Condone correct points given as position vectors. NMS: Correct answer scores 2/2. NMS: One correct invariant point and only one incorrect point scores SC1.	AO1.1b	A1	$x = -3y$ $\therefore$ two invariant points are (0, 0) and (−3, 1)
			Total 2 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$y = 0$
			Total 1 mark

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds $k = 2$	AO1.1b	B1	$k - 2 = 0 \Rightarrow k = 2$
(b)	States correct transformation	AO1.2	B1	Reflection in the y-axis
(c)(i)	Finds product **BA** Allow one slip	AO1.1a	M1	$\mathbf{BA} = \begin{bmatrix} -1 & -2 \\ 1 & k \end{bmatrix}$
	Obtains inverse FT ‘their’ **BA** provided M1 awarded	AO1.1b	A1F	$(\mathbf{BA})^{-1} = \dfrac{1}{-k-(-2)}\begin{bmatrix} k & 2 \\ -1 & -1 \end{bmatrix}$

	Finds $\mathbf{A}^{-1}$ and $\mathbf{B}^{-1}$	AO1.1b	B1	$\mathbf{A}^{-1} = \frac{1}{k-2}\begin{bmatrix} k & -2 \\ -1 & 1 \end{bmatrix}$ $\mathbf{B}^{-1} = \frac{1}{-1}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $\mathbf{A}^{-1}\mathbf{B}^{-1} = \frac{1}{(k-2)\times(-1)}$ $\begin{bmatrix} k+0 & (-2\times-1)+0 \\ (-1\times1)+0 & 0+(-1\times1) \end{bmatrix}$ That is the same as $(\mathbf{BA})^{-1}$ $(\mathbf{BA})^{-1} = \frac{1}{2-k}\begin{bmatrix} k & 2 \\ -1 & -1 \end{bmatrix} = \mathbf{A}^{-1}\mathbf{B}^{-1}$
	Obtains correct $\mathbf{A}^{-}$, $\mathbf{B}^{-}$, and shows that $(k-2)\times(-1) = 2-k = -k-(-2)$ thus completing verification Must clearly show $\mathbf{A}^{-}$, × $\mathbf{B}^{-}$, method for this mark − disallow if answer simply stated	AO2.1	R1	
(ii)	Uses equation for identity from definition	AO3.1a	M1	We require to demonstrate that: $(\mathbf{NM}) \times \{\mathbf{M}^{-1}\mathbf{N}^{-1}\} = I$
	Commences argument by manipulating the matrix products within the equation with clear pairing	AO2.1	R1	$(\mathbf{NM}) \times \mathbf{M}^{-1}\mathbf{N}^{-1} = \mathbf{N}(\mathbf{M} \times \mathbf{M}^{-1})\mathbf{N}^{-1}$
	Clearly demonstrates that $\mathbf{M} \times \mathbf{M}^{-1} = I$ used	AO2.4	B1	$= \mathbf{N}\,I\,\mathbf{N}^{-1}$ $= \mathbf{N}\mathbf{N}^{-1}$
	Completes the argument using rigorous reasoning with definition of matrix inverse and associativity mentioned Must see all working with correct pairing of each matrix with inverse	AO2.1	R1	$= I$ Using definition of matrix inverse and associativity of matrix multiplication Hence true for all non-singular matrices **N** and **M**
				Total 10 marks

Q4.

(a) (i) $\mathbf{X}^2 = \begin{bmatrix} 7 & 2 \\ 3 & 6 \end{bmatrix}$; $(m=)7$

(m =)7 or 7 as top left element of $\boldsymbol{X}^2$

B1

1

(ii) $\mathbf{X}^3 = \begin{bmatrix} 13 & 14 \\ 21 & 6 \end{bmatrix}$;

At least 2 elements correct

M1

$7\mathbf{X} = \begin{bmatrix} 7 & 14 \\ 21 & 0 \end{bmatrix}$

PI

B1

$\mathbf{X}^3 - 7\mathbf{X} = \begin{bmatrix} 13-7 & 14-14 \\ 21-21 & 6-0 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$

Ft on c's m value

A1F

$= 6\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 6\mathbf{I}$

CSO Accept either form but at least one must be shown explicitly

A1

4

(b) (i) Reflection in the x-axis

OE

B1

1

(ii) $\mathbf{B} = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

Either OE. For M mark, accept dec. equiv. (at least 3sf) for $\frac{1}{\sqrt{2}}$

M1

$= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

NMS SC1 for $k = \frac{1}{\sqrt{2}}$ *or better.*

A1

2

(iii) **AB** $\begin{bmatrix} -1 \\ 2 \end{bmatrix} = k\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

Attempt to find ***AB*** $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

M1

$= k\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}\begin{bmatrix}-3\\1\end{bmatrix}$ $\left\{\text{or } k\begin{bmatrix}1 & -1\\-1 & -1\end{bmatrix}\begin{bmatrix}-1\\2\end{bmatrix}\right\}$

Either $\begin{bmatrix}1 & -1\\1 & 1\end{bmatrix}\begin{bmatrix}-1\\2\end{bmatrix}=\begin{bmatrix}-3\\1\end{bmatrix}$ or $\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}\begin{bmatrix}1 & -1\\1 & 1\end{bmatrix}=\begin{bmatrix}1 & -1\\-1 & -1\end{bmatrix}$

A1

$= k\begin{bmatrix}-3\\-1\end{bmatrix}$

Completing the matrix mult. to reach a 2 × 1 matrix

m1

(Image of P is the point) $\left(-\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$

CSO

SC *Wrong order, works with* ***BA*** $\begin{bmatrix}-1\\2\end{bmatrix}$*,*
mark out of a max of M1A0 m1A0

A1

4

[12]

Q5.

(a) $\begin{bmatrix}1 & 0\\0 & 3\end{bmatrix}$

Matrix in form $\begin{bmatrix}\lambda & 0\\0 & \mu\end{bmatrix}$, *where* $\lambda \neq 0$, $\mu \neq 0$ *and* $\lambda \neq \mu$

M1

$\begin{bmatrix}1 & 0\\0 & 3\end{bmatrix}$

A1

2

(b) (i) $y = \sqrt{3}\,x = \tan 60°\,x$ $\begin{bmatrix}\cos 120° & \sin 120°\\\sin 120° & -\cos 120°\end{bmatrix}$

$\begin{bmatrix}\cos 120 & \sin 120\\\sin 120 & -\cos 120\end{bmatrix}$ PI

For M mark, condone dec approx 0.86 or 0.87 or better in place of sin120°

M1

Required matrix is $\begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

OE but must be in exact surd form.

A1

2

(ii) $\begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ =......

*Attempt to multiply c's (b)(i) 2 by 2 matrix and c's (a) 2 by 2 matrix in **correct** order.*

M1

$= \begin{bmatrix} -\frac{1}{2} & \frac{3\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{3}{2} \end{bmatrix}$

OE but must be in exact surd form.

A1

2

[6]

Q6.

(a) Determinant of matrix = −8 + 9 = 1

Finding determinant and multiplying by area

M1

Area = 3 × 1 = 3 (square units)

CAO − must show multiplication or refer to scale factor / invariant area or equivalent

A1

2

(b) (i) $\begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix}\begin{bmatrix} x \\ mx+c \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$

⇒

(x') = 4x + 3(mx + c)
(y') = − 3x − 2(mx + c)

x', y' in terms of x, y, m, c

M1

Invariant lines ⇒ $y' = mx' + c$

$\Rightarrow -3x - 2mx - 2c = 4mx + 3m^2 x + 3mc + c$

$\Rightarrow 0 = (3m^2 + 6m + 3)x + 3mc + 3c$

Use of $y' = mx' + c$

A1

$\Rightarrow 3m^2 + 6m + 3 = 0 \quad 3mc + 3c = 0$

Attempt at solving equations where coefficients = 0 or compares coefficients

M1

$3(m + 1)^2 = 0 \qquad 3c(m + 1) = 0$

$\Rightarrow m = -1 \qquad c$ can be any value

Finding the correct value of m

A1

$\Rightarrow$ lines are $y = -x + c$

Fully correct line − no restriction on c

A1

5

SPECIAL CASES

$$\begin{pmatrix} 4 & 3 \\ -3 & -2 \end{pmatrix}\begin{pmatrix} x \\ -x+c \end{pmatrix} = \begin{pmatrix} x+3c \\ -x-2c \end{pmatrix}$$

$x' = x + 3c$

$y' = -x - 2c$

SC1 − Correct multiplication as shown

Consider

$-x' + c$
$= -(x + 3c) + c$
$= -x - 3c + c$
$= -x - 2c$
$= y'$

Hence $y = -x + c$ is an invariant line

SC2 − correct multiplication as shown above and full algebraic solution using $y' = -x' + c$

(ii) When $c = 0$, $y = -x$ is a line of invariant points

Any equivalent form

B1

1

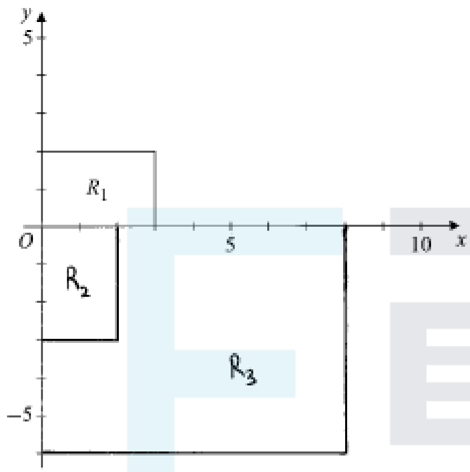
[8]

Q7.

(a) (i) *Rectangle with vertices (0, 0), (0, –3), (2, –3), (2, 0)*

B1

1



(ii)

Rectangle with vertices ***either*** *whose x-coords are c's (a)(i) x-coord vertices multiplied by 4 or whose y-coords are c's (a)(i) y-coord vertices multiplied by 2*

M1

A2 if rectangle with vertices (0, 0), (0, –6), (8, –6), (8, 0) (A1 if either the four x-coords are correct or the four y-coords are correct)

A2,1

3

(b) (i) Matrix is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

B1

1

(ii) $\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} =$

Attempt to multiply $\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$ *with c's (b)(i) matrix in* ***either*** *order.*

M1

Multiplication in ***correct*** *order with at least two of the four ft multiplications carried out correctly.*

m1

$\begin{bmatrix} 0 & 4 \\ -2 & 0 \end{bmatrix}$

For $\begin{bmatrix} 0 & 4 \\ -2 & 0 \end{bmatrix}$

NMS $\begin{bmatrix} 0 & 4 \\ -2 & 0 \end{bmatrix}$ *scores B3*

$\begin{bmatrix} 0 & 2 \\ -4 & 0 \end{bmatrix}$ *scores B1*

A1

3

[8]

Q8.

(a) $\begin{bmatrix} c & s \\ -s & c \end{bmatrix}$

B1

$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}\begin{bmatrix} X \\ Y \end{bmatrix}$

M1

$= \begin{bmatrix} cX + sY \\ -sX + cY \end{bmatrix}$

A1

3

(b) (i) $(cX + sY)^2 - 6(cX + sY)(-sX + cY) - 7(-sX + cY)^2 = 8$

$(c^2X^2 + 2csXY + s^2Y^2) - 6([c^2 - s^2]XY) + sc\{Y^2 - X^2\})$
$- 7(s^2X^2 - 2csXY + c^2Y^2) = 8$

Substn. for x & y in eqn. and multiplying out

M1

$p = c^2 + 6sc - 7s^2$

A1

$q = 16cs - 6(c^2 - s^2)$

AG

A1

$r = s^2 - 6sc - 7c^2$

A1

(ii) Factorising:

$3s^2 + 8sc - 3c^2 = (3s - c)(s + 3c) = 0$

Or by double angles

M1A1

Deducing a tan value

M1

$\tan\theta = \frac{1}{3}$ (θ acute)

A1

$\cos\theta = \frac{3}{\sqrt{10}}$, $\sin\theta = \frac{1}{\sqrt{10}}$

Both

A1

Subst^g. sensible values back for p and r

M1

$2X^2 - 8Y^2 = 8$

A1

$\frac{X^2}{2^2} - \frac{Y^2}{1^2} = 1$

CSO

A1 **8**

(iii) Since C' is a hyperbola, and it is just C rotated, it follows that C is a hyperbola

E1 **1**

[16]

Q9.

(a)

$$\begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

If A1 not scored, award M1A0 for all correct entries expressed in trig form eg

M1

$$\begin{bmatrix} \cos 135 & -\sin 135 \\ \sin 135 & \cos 135 \end{bmatrix}$$

A1

2

(b) (i) **M** = $\begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} = \sqrt{2} \times \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \left(= \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \right)$

Or better
PI by cand. having both a correct scale factor of enlargement and a correct corresponding angle of rotation

M1

Scale factor of enlargement is $\sqrt{2}$

SF = $\sqrt{2}$ *OE surd form*

A1

Angle of rotation is 135 (degrees anticlockwise)

Angle = 135 OE eg −225

If M0 give B1 for SF = $\sqrt{2}$ *OE surd and*

B1 for angle = 135 OE

A1

3

(ii) For $\mathbf{M}^2$, SF of enlargement = 2

OE If incorrect, ft on [c's SF in (b)(i)]²

OE, eg −90(degrees), eg 90 (degrees) clockwise

B1F

Angle of rotation is 270 (degrees anticlockwise)

If incorrect, ft on 2 × c's angle in (b)(i)

(neither B1F B1 nor B1 B1F is possible)

B1F

2

(iii) $\mathbf{M}^2 = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$

For complete method (matrix calculation or geometrical reasoning)

$\mathbf{M}^4 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix}$

Matrix for $\mathbf{M}^2$ *could be seen earlier*
(M0 if >1 independent error in matrix multiplication)

Geometrically SF = 4, rotation angle= 540
OE scores M1 and completion scores A1

M1

$\mathbf{M}^4 - 4\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = -4\mathbf{I}$

Either of these two forms convincingly shown

A1

2

(iv) $\mathbf{M}^{2012} = (\mathbf{M}^4)^{503} = (-4\mathbf{I})^{503} = -(2^2)^{503}\mathbf{I} = -2^{1006}\,\mathbf{I}$

OE Fully explained, algebraically from $(-4\mathbf{I})^{503}$, or geometrically

E1

$\mathbf{M}^{2012} = -2^{1006}\mathbf{I}$

$\mathbf{M}^{2012} = -2^{1006}\mathbf{I}$ $(n = 1006)$ (B0 if FIW)

B1

Geometrically: $\mathbf{M}^{2012}$ represents an enlargement with SF 2^{1006} followed by a rotation of angle 2012 × 135° ie 754.5 revolutions, being equivalent to rotation of 180° ie matrix is $-\mathbf{I}$ so $\mathbf{M}^{2012} = -2^{1006}\,\mathbf{I}$)

2

[11]

Q10.

(a) (i) $\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$

B1

1

(ii) $\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$

B1

1

(b) (i) $\mathbf{AB} = \begin{bmatrix}-20 & 14\\ 14 & -10\end{bmatrix}$

M1A0 if 3 entries correct

M1A1

2

(ii) $\mathbf{A} + \mathbf{B} = \begin{bmatrix}0 & 5\\ -5 & 0\end{bmatrix}$

B1

$$(\mathbf{A}+\mathbf{B})^2 = \begin{bmatrix} -25 & 0 \\ 0 & -25 \end{bmatrix}$$

B1

... = −25**I**

*ft if c's (**A** + **B**)^2 is of the form k**I***

B1F

3

(c) (i) Rot'n 90° clockwise, enlargem't SF 5

OE

B2, 1

2

(ii) Rotation 180°, enlargement SF 25

Accept 'enlargement SF −25';

ft wrong value of k

B2, 1F

2

(iii) Enlargement SF 625

B1 for pure enlargement; ft ditto

B2, 1F

2

[13]

Q11.

(a) Det(**M**) = – 1

B1

Magnitude = 1 $\Rightarrow$ area invariant

FT area s.f.

B1√

–ve sign $\Rightarrow$ cyclic order of vertices is reversed
OR "reflection" involved

B1

3

(b) <u>**Method 1**</u>

Char. Eqn.: $\lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$

Finding and solving attempt

M1 A1

Subst^g. back: $\lambda = 1 \Rightarrow y = \frac{1}{2}x$

M1 A1

and $\lambda = -1$ $\Rightarrow y = \frac{1}{4}x$

A1

5

Method 2

$$\begin{bmatrix} -3 & 8 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} x \\ mx \end{bmatrix} = \begin{bmatrix} (8m-3)x \\ (3m-1)x \end{bmatrix}$$

Attempted

(M1)

Use of $y' = mx'$: $3m - 1 = 8m^2 - 3m$

(M1)

Solving a quadratic eqn. in m = $\frac{1}{4}$, $\frac{1}{2}$

From $(4m - 1)(2m - 1) = 0$

(M1A1)

$p = \frac{1}{2}$ and $q = \frac{1}{4}$

(A1)

(c) $p = \frac{1}{2} = \tan\theta$

$\Rightarrow \cos 2\theta = \frac{3}{5}$ and $\sin 2\theta = \frac{4}{5}$

For these attempted and used in a reflection matrix

M1

$$\mathbf{R} = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{bmatrix}$$

A1

2

[10]

Q12.

(a) (i) $\mathbf{A}^2 = \begin{bmatrix} -2 & 2\sqrt{3} \\ -2\sqrt{3} & -2 \end{bmatrix}$

M1 if at least two entries correct

M1A1 **2**

(ii) $\mathbf{A}^3 = \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}$

if at least two entries correct

M1

……..= 8**I**

A1 **2**

(b) (i) $\mathbf{A}^3$ gives enlargement with SF 8 (centre the origin)

M1 for enlargement (only);

ft wrong value for k

M1A1F **2**

(ii) Enlargement and rotation

Some detail needed

M1

Enlargement scale factor 2

A1

Rotation through 120° (antic'wise)

A1 **3**

[9]

Q13.

$\begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}$ & $\begin{bmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{bmatrix}$

used or written down

B1

Mult'n of these in the correct order

at least two entries correct

B1

Use of addition formulae

At least once

M1

$$\begin{bmatrix} \cos(2\alpha+\beta) & \sin(2\alpha+\beta \\ \sin(2\alpha+\beta & -\cos(2\alpha+\beta \end{bmatrix}$$

ft only for use of clockwise rot'n and/or mult'n in wrong order

A1F

Reflection ...

ft as above

A1F

... in $y = x \tan\left(\alpha + \frac{1}{2}\beta\right)$

ft as above

A1F

[6]

Q14.

(a) (i) $\mathbf{r} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}$

Any correct vector line equation; B1 if one vector correct, or if both correct but equation not in correct form

B2,1

2

(ii) $\mathbf{r} = \begin{bmatrix} 1 & 4 & -3 \\ 2 & -1 & 0 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1+2\lambda \\ 2+3\lambda \\ 3+6\lambda \end{bmatrix} = \begin{bmatrix} -4\lambda \\ \lambda \\ -\lambda \end{bmatrix}$

Attempt at multiplication

M1

At least one entry correct

A1

All three correct

A1

Clear and valid explanation that this is a line through O

E1 4

(b) (i) $\begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} p \\ \frac{1}{2}p+k \end{bmatrix} = \begin{bmatrix} 3p+4k \\ \frac{3}{2}p-k \end{bmatrix}$

For LHS

B1

For RHS

M1A1

Answer satisfies $y = \frac{1}{2}x - 3k$

A1 4

(ii) Equal gradients, hence parallel

ft if previous answer is of the form $y = \frac{1}{2}x + c$

E1F

Distance = $|k - c| \cos\theta$ with $\tan\theta = \frac{1}{2}$

Allow incorrect value of c here

M1

$\ldots = \frac{8k}{\sqrt{5}}$

Allow 3.58k

A1 3

[13]

Q15.

(a) (i) Coords (3, 2), (9, 2), (9, 4), (3, 4)

M1 for multn of x by 3 or y by 2 (PI)

M1A1 2

(ii) R_2 shown correctly on insert

B1 1

(b) (i) R_3 shown correctly on insert

B1 for rectangle with 2 vertices correct;
ft if c's R_2 is a rectangle in 1st quad

B2,1F
2

(ii) Matrix of rotation is $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

B1
1

(c) Multiplication of matrices
(either way)
or other complete method

M1

Required matrix is $\begin{bmatrix} 0 & 2 \\ -3 & 0 \end{bmatrix}$

A1
2
[8]

Q16.

(a) 1

B1

(b) –1

B1

(c) 9

B1
[3]

Q17.

(a) $\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}\begin{bmatrix} x \\ 2x+1 \end{bmatrix} = \begin{bmatrix} x-1 \\ -x+2 \end{bmatrix}$

Use of $\begin{bmatrix} x \\ 2x+1 \end{bmatrix}$*; either term correct*

M1A1

giving $y = 1 - x$
CSO

A1

3

(b) Finding det(**B**) (= 1) and mult^g. by 4.5

M1

= 4.5 cm^2

ft

A1

2

(c) $\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}\begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

AB attempted

M1A1

A1

This is a shear

M1

parallel to the x-axis

A1

mapping (eg) (1, 1) to (3, 1)

Any point not on Ox and its image

A1

5

[10]

Q18.

(a) Rotation 45° (anticlockwise) (about O)

M1 for 'rotation'

M1A1

2

(b) Reflection in $y = x \tan 22.5°$

M1 for 'reflection'

M1A1

2

(c) Rotation 90° (anticlockwise) (about O)

M1 for 'rotation' or correct matrix;
ft wrong angle in (a)

M1A1F

2

(d) Identity transformation

ft wrong mirror line in (b); B1 for $\mathbf{B}^2 = \mathbf{I}$

B2,1F

2

(e) $\mathbf{AB} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

allow M1 if two entries correct

M1A1

Reflection in $y = x$

A1

3

[11]

Q19.

(a) (i) $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 0 & 2k \\ 2k & 0 \end{bmatrix}$

B1

1

(ii) $\mathbf{A}^2 = \begin{bmatrix} 2k^2 & 0 \\ 0 & 2k^2 \end{bmatrix}$

B1 if three entries correct

B2,1

2

(b) $(\mathbf{A} + \mathbf{B})^2 = \begin{bmatrix} 4k^2 & 0 \\ 0 & 4k^2 \end{bmatrix}$

B1 if three entries correct

B2,1

$\mathbf{B}^2 = \mathbf{A}^2$, hence result

B1B1

4

(c) (i) $\mathbf{A}^2$ is an enlargement (centre O)

M1

with SF 2

Condone $2k^2$

A1

2

(ii) Scale factor is now $\sqrt{2}$

Condone 2k

B1

Mirror line is $y = x\tan 22\frac{1}{2}^{\circ}$

M1 A1

3

[12]

Q20.

(a) (i) $\frac{1}{5}\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$

1/det

B1

Transposed matx. of cofactors

B1

2

(ii) $\begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{A}^{-1}\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} \frac{1}{5}(X+2Y) \\ \frac{1}{5}(Y-2X) \end{bmatrix}$

M1

ft

A1F

2

(b) **A** = $\sqrt{5}\begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$

B1

Enlargement sf $\sqrt{5}$ (centre O)

Two components in any order

M1A1

+ Rotation thro' $\cos^{-1}(1/\sqrt{5})$

or 63.4° or 1.11 rads

M1A1

5

(c) (i) $p = ½$, $q = 3$ noted

Or form $\dfrac{x^2}{\frac{1}{2}} + \dfrac{y^2}{3} = 1$ *shown*

B1

1

(ii) $6x^2 + y^2 = 3 \Rightarrow$

$\frac{6}{25}\left(X^2 + 4XY + 4Y^2\right) + \frac{1}{25}\left(Y^2 - 4XY + 4X^2\right) = 3$

Substg. for x and y

M1

$\Rightarrow 10X^2 + 20XY + 25Y^2 = 75$

$\Rightarrow 2X^2 + 4XY + 5Y^2 = 15$

ANSWER GIVEN

A1

2

(iii) It is just an enlarged rotation of E, hence still an ellipse

B1

1

[13]

Q21.

(a) (i) Matrix is $\begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}$

M1 for $\begin{bmatrix} \cos 30° & \sin 30° \\ -\sin 30° & \cos 30° \end{bmatrix}$ *(PI)*

M1A1

2

(ii) Matrix is $\begin{bmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix}$

M1 for $\begin{bmatrix} \cos 60° & \sin 60° \\ \sin 60° & -\cos 60° \end{bmatrix}$ *(PI)*

M1A1

(b) SF 2, line $y = \dfrac{1}{\sqrt{3}}x$

OE

B1B1

2

(c) Attempt at **BA** or **AB**

M1

BA = $\begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}$

m1 if zeros in correct positions

m1A1

Enlargement SF 4

*ft use of **AB** (answer still 4)*

*or after **BA** =* $\begin{bmatrix} 0 & k \\ k & 0 \end{bmatrix}$

B1F

... and reflection in line $y = x$

*ft only from **BA** =* $\begin{bmatrix} 0 & k \\ k & 0 \end{bmatrix}$

B1F

5

[11]

Q22.

(a) (i) $\mathbf{M}^2 = \begin{bmatrix} 12 & 0 \\ 0 & 12 \end{bmatrix}$

M1 if zeroes appear in the right places

M1A1

= 12**I**

ft provided of right form

A1F

3

(ii) $q\cos 60° = \frac{1}{2}q = \sqrt{3} \Rightarrow q = 2\sqrt{3}$

*OE **SC** $q = 2\sqrt{3}$ NMS 1/3*

M1A1

Other entries verified

surd for sin 60º needed

E1

3

(b) (i) SF = $q = 2\sqrt{3}$

ft wrong value for q

B1F

1

(ii) Equation is $y = x \tan 30^{\circ}$

B1

1

(c) $\mathbf{M}^4 = 144\mathbf{I}$

PI; ft wrong value in (a)(i)

B1F

$\mathbf{M}^4$ gives enlargement SF 144

ft if c's $\mathbf{M}^4 = k\mathbf{I}$

B1F

2

[10]

Q23.

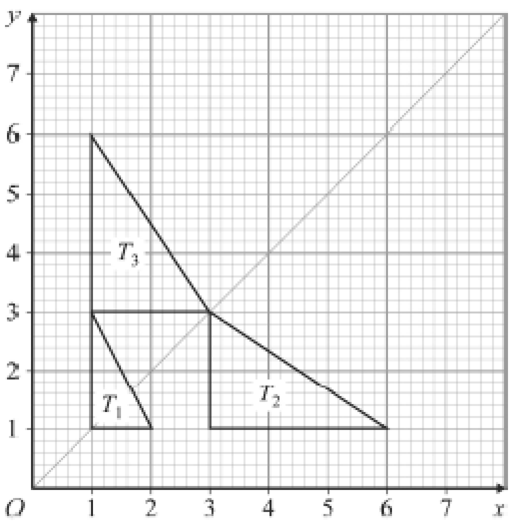
(a) Matrix is $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

M1 if zeros in correct positions; allow NMS

M1A1

2

(b)



Third triangle shown correctly

M1A0 if one point wrong

M1A1

2

(c) Matrix of reflection is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Alt: calculating matrix from the coordinates: M1 A2,1

B1

Multiplication of above matrices

in correct order

M1

Answer is $\begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$

*ft wrong answer to (a); **NMS** 1/3*

A1F

3

[7]

Q24.

(a) (i) **A + B** = $\begin{bmatrix} \sqrt{3} & 0 \\ 1 & 0 \end{bmatrix}$

M1A0 if 3 entries correct;

Condone $\frac{2\sqrt{3}}{2}$ *for* $\sqrt{3}$

M1A1

2

(ii) **BA** = $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Deduct one for each error;
SC** B2,1 for **AB

B3,2,1

3

(b) (i) Rotation 30° anticlockwise (abt O)

M1 for rotation

M1A1

2

(ii) Reflection in $y = (\tan 15°)x$

M1 for reflection

M1A1

2

(iii) Reflection in x-axis

1/2 for reflection in y-axis
*ft (M1A1) only for the **SC***

B2F

2

Alt: Answer to (i) followed by answer to (ii)

M1A0 if in wrong order
or if order not made clear

M1A1F

(2)

[11]

Q25.

(a) $\mathbf{M} = \begin{bmatrix} 0 & -3 \\ -3 & 0 \end{bmatrix}$

B1 if subtracted the wrong way round

B2,1

2

(b) $p = 3$

ft after B1 in (a)

B1F

L is $y = -x$

Allow $p = -3$, L is $y = x$

B1

2

(c) $\mathbf{M}^2 = \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix}$

Or by geometrical reasoning; ft as before

B1F

... = 9**I**

ft as before

B1F

2

[6]

Q26.

(a) (i) Reflection ...

M1

... in $y = -x$
OE

A1
2

(ii) $\mathbf{A}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
M1A0 for three correct entries

M1A1
2

(iii) $\mathbf{A}^2 = \mathbf{I}$ or geometrical reasoning

E1
1

(b) (i) $\mathbf{B}^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
M1A0 for three correct entries

M1A1

$\boldsymbol{b}^2 - \mathbf{A}^2 = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$
ft errors, dependent on both M marks

A1ft
3

(ii) $(\mathbf{B} + \mathbf{A})(\mathbf{B} - \mathbf{A}) = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$

B1

$\ldots = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$
ft one error; M1A0 for three correct (ft) entries

M1
A1$\checkmark$
3

[11]

Q27.

(a) For an *invariant line*, all points on the line have image points also on the line.

For a line of invariant points, all points on the line map onto themselves.
Clearly explained

B1 1

(b) Det = 3

B1

The s.f.. of area enlargement under T

Must mention area

B1 2

(c) Setting $x' = x$ and $y' = y$ and solving

$x = 2x - y$ *and* $y = -x + 2y$

M1

$\Rightarrow y = x$

A1

Or

Char. Eqn. is $\lambda^2 - 4\lambda + 3 = 0$

M1

$\lambda = 1$ gives l.o.i.p.s and $y = x$

A1 2

(d) (i) $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} x \\ -x+c \end{bmatrix}$

M1

$= \begin{bmatrix} 3x - c \\ -3x + 2c \end{bmatrix}$

A1 2

(ii) $y' = -x' + c$ also

M1

$\Rightarrow y = -x + c$ invariant for all c

A1 2

(e) Stretch, s.f. 3,

M1A1

perp^r. to $y = x$ (or || to $y = -x$)

A1

3

[12]

Q28.

(a) (i) $M^2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

M1 if 2 entries correct

M1

M1A1 if 3 entries correct

A2,1

3

(ii) $M^4 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

ft error in $\mathbf{M}^2$ provided no surds in $\mathbf{M}^2$

B1ft

1

(b) Rotation (about the origin)

M1

... through 45° clockwise

A1

2

(c) Awareness of $\mathbf{M}^8 = \mathbf{I}$

OE; NMS 2/3

M1

$M^{2006} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

complete valid method

m1

ft error in $\mathbf{M}^2$ as above

A1ft

3

[9]

Q29.

(a) Transformation is a reflection in $y = x$

B2

2

(b) Matrix is $\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

M1 for matrix for a rotation;

M1

A1 for correct trig expressions

A2,1

3

(c) *Attempt to multiply the matrices*

M1

... in the correct order

m1

Matrix is $\begin{bmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

Wrong answer to (b)

A1ft

3

[8]

Q30.

(a) (i) *D (4, 0)*

E (8, –4), F (10, –2)

M1 if at least one point correct

M1A1
A1A1

4

(ii) Correct sketch

Ft one error

m1A1F

2

(b) (i) Scale factor is $2\sqrt{2}$

NMS 2/2; 1/2 for AWRT 2.8

M1A1

2

(ii) Angle 45°

NMS 2/2; condone 315°; 1/2 for AWRT 44-46° OE

M1A1

2

[10]

Q31.

(a) $\begin{vmatrix} 6.4 & -7.2 \\ -7.2 & 10.6 \end{vmatrix} = 67.84 - 51.84 = 16$

Attempt at det.

M1A1

2

(b) Inv. pts.g.b. $x' = x$, $y' = y$

B1

Subst^g. in eqns. $6.4x - 7.2y = x$

$-7.2x + 10.6y = y$

M2

A1

$y = \frac{3}{4}x$

A1

5

Alt. I

Char. Eqn. is $\lambda^2 - 17\lambda + 16 = 0$

M1A1

$\lambda = 1$ or 16

A1

$\lambda = 1$ for l.o.i.p.s $\Rightarrow 5.4x - 7.2y = 0$

i.e. $y = \frac{3}{4}x$

ignore $\lambda = 16$ work

M1A1

(5)

Alt. II

$y = mx$ a l.o.i.p.s

$$\begin{bmatrix} 6.4 & -7.2 \\ -7.2 & 10.6 \end{bmatrix}\begin{bmatrix} x \\ mx \end{bmatrix} = \begin{bmatrix} 6.4x - 7.2mx \\ -7.2x + 10.6mx \end{bmatrix}$$

B1

$-7.2x + 10.6mx = m(6.4x - 7.2mx)$ also

M1
A1

$\Rightarrow 7.2m^2 + 4.2m - 7.2 = 0$

$\Rightarrow (4m - 3)(3m + 4) = 0$

A1

$\Rightarrow y = \frac{3}{4}x$ or $y = -\frac{4}{3}x$

A1

Checking which one works

B1

(5)

[7]

Examiner reports

Q1.

The vast majority of students realised that $\mathbf{M}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$, and were able to multiply out to produce two equivalent equations. However, a significant number of these students were unable to use their equation(s) to find an invariant point. Many chose to solve them as simultaneous equations, and either made an error or confused themselves when every term cancelled out. A minority of students knew that (0, 0) would be invariant and gave this as the only possible point.

Q4.

In this question on matrices, parts (a)(i) and (a)(ii) were almost always answered correctly, although some students left their answer to (a)(ii) in the form $\begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$ without indicating that they clearly appreciated what **I** represented. A correct description of the geometrical transformation represented by **A** was usually given although reflection in the line $y = x$ was the most frequent incorrect answer.

Most students scored the method mark in part (b)(ii) but some then gave the incorrect value $\sqrt{2}$ for k. Part (b)(iii) proved to be more of a challenge. A significant minority of students could not make any real progress. Most realised that the matrices **A** and **B** needed to be multiplied but the product was not always linked with the matrix for **P**,$\begin{bmatrix} -1 \\ 2 \end{bmatrix}$, in a correct manner. Some better approaches just made the mistake of using **BA** instead of **AB**. For the final mark the examiners wanted the coordinates of the image of **P** to be written in coordinate form rather than left as a matrix.

Q5.

This question, which tested matrix transformations, was not attempted by a significantly higher proportion of students than the other questions on the paper. However, there were more students who scored full marks for this question than for any of the last seven questions on the paper. Part (a) was generally answered correctly with many able to just write down the correct matrix $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ for the stretch. The most common wrong answers were either $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$ or $\begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix}$. In part (b)(i), the majority of students quoted or used the correct matrix from the formulae booklet but some took θ to be 30° instead of 60°. A small minority left their answer in trigonometric form instead of using the surd form as instructed. In part (b)(ii) there were a significant number of students who wrote down and multiplied their matrices in the wrong order.

Q6.

This question proved to be the most demanding. Many students knew what to do for part (a) but often failed to show all working out leading to a loss of marks. Part (b)(i) proved elusive for all but the most able students. When asked to find all invariant lines the most efficient and valid method involves obtaining y' and x' in terms of x, y and c and then trying to make them fit the equation $y' = mx' + c$. This method was rarely seen and yet is the correct standard approach. Other approaches such as using eigenvalues are more appropriate to lines of invariant points only.

Q7.

Parts (a)(i) and (a)(ii) were generally answered correctly although some rectangles for R_3 were drawn which had vertices with either the wrong x-coordinates or the wrong y-coordinates.

The most common wrong answer for part (b)(i) was the matrix which represented a rotation of 90° anticlockwise and in part (b)(ii) it was not uncommon to see the matrices being multiplied in the wrong order. A small number of candidates gave solutions which had matrices left in terms of trigonometric functions; these were penalised.

Q8.

Although there were a few excellent, complete responses to this question, in general it turned out to reflect the lack of preparedness on the part of candidates for the demands presented to them here. The three marks in part (a) were readily and easily acquired by most candidates attempting it. However, 156 candidates were unable to proceed beyond the given result of part (a), and made no attempt at the remaining parts of this question.

In part (b)(i), it was not uncommon to find that candidates were less than comfortable with squaring out their x and y terms in the new variables X and Y, and collecting them up proved to be far more problematical than should have been the case. Rather a lot of candidates seemed to think that the XY term came exclusively from the xy term, and such efforts foundered at this point.

Few candidates got to the correct equation of the hyperbola. A suitable answer to part (b)(iii) required only an awareness that the original curve had been rotated into the final hyperbola, but only a few candidates answered it without having worked their way to it. Of these, most offered spurious reasons, such as the not uncommon reference to the suggestion that the original curve was a hyperbola because this final one was an ellipse.

Q9.

This was the least well-answered question on the paper with only the most able scoring full marks. Part (a) was generally well-answered with only a small minority failing to give the matrix in surd form. In part (b)(i), better students showed clear working which related to part (a), and stated correct answers for both the scale factor of the enlargement and the angle of the rotation.

In part (b)(ii) the most common method seen was to find the matrix $\mathbf{M}^2$ first and use it to find the scale factor and angle rather than stating the scale factor to be the square of the scale factor in part (b)(i) and the angle to be double the angle in part (b)(i). Those using the matrix $\mathbf{M}^2$ usually gave the correct scale factor but 90° anticlockwise was a common wrong angle.
Students generally scored both marks in part (b)(iii), although some others made an arithmetic error in their matrix multiplication or did not complete the question appropriately and thus lost the final mark. Part (b)(iv) proved to be very challenging and only a small number showed the deduction sufficiently rigorously. There were many examples of students obtaining the correct value of n but from incorrect reasoning. Most attempted this part using algebraic methods but relatively few gave due consideration to the negative sign.

Some excellent deductions based on geometric reasoning using the earlier results in the question were also seen.

Q10.

Like the preceding questions, this one produced a good opportunity for most candidates to score high marks. The first two marks were sometimes lost because the candidate failed to provide numerical values for the sines and cosines in their matrices. Also the first matrix was often that of a 90° anticlockwise rotation, rather than a clockwise one as required. Most candidates earned two marks in part (b)(i), relatively few finding **BA** instead of **AB**. Full marks were very common in part (b)(ii). The geometrical interpretations asked for in part (c) were mostly correct, though some floundered somewhat in the first part. In part (c)(ii), the answer 'enlargement with scale-factor −25' was acceptable, and very common.

Q11.

In part (a), many candidates realised the significance of both the sign and magnitude of the determinant of the transformation matrix, although there are still far too many who try to describe area scale factors without using the key word "area".

In part (b), almost all attempts managed to find or verify the given value of p, but most who did go on to try and find a value for q as well simply did what amounted to the same working again. The more successful bids found the eigenvalues first and then deduced p and q.

Attempts at part (c) fell almost equally into the three categories of correct, incorrect – usually a 45° or 90° rotation matrix – or nothing at all.

Q12.

The first matrix calculation in this question involved a fair number of surds and minus signs, and yet it was very accurately performed by the majority of candidates. After that, it was relatively easy to reach a matrix of the required form in part (a)(ii). Almost all candidates knew how to describe the transformation represented by this matrix, but by contrast many had a long and unsuccessful struggle to describe the transformation required in part (b)(ii). An acceptable alternative here was an enlargement with scale factor –2 and a rotation through 60° clockwise.

Q13.

This question was not answered at all well by many candidates. Some failed to write down the matrices correctly and some failed to carry out the required matrix multiplication with the matrices in the right order. Then, relatively few candidates spotted the need to use the addition formulae, though those who did were mostly successful in giving the correct geometrical interpretation of the resulting matrix.

Q14.

In part (a)(i), most candidates knew how to convert the given equations into a suitable vector equation, but some lost a mark by writing 'L = ...' rather than '**r** = ...'. Part (a)(ii) was likewise answered well but often not perfectly, the usual fault here being a failure to show clearly why the line must pass through the origin.

Part (b) was found more difficult than part (a), even though it was in a two-dimensional context. Much confusion arose from the use of the same letters x and y to denote the coordinates of a general point before and after the transformation. Many candidates were able to carry out an appropriate matrix multiplication but were unsuccessful in their attempts to form a linear equation for the image line. In part (b)(ii), most candidates earned the first mark by comparing the gradients of the two lines, even if the equation of the second line was incorrect, but very few indeed realised that the 'distance' between the two parallel lines referred to the perpendicular distance between them.

Q15.

As the candidates turned the page to tackle this question, there was a noticeable dip in the level of their performance. The majority of candidates showed a surprising lack of ability to work out the coordinates of image points under a transformation given by a matrix. A common misunderstanding was to carry out a two-way stretch with centre (1, 1) instead of with centre (0, 0). Luckily the candidates still had the chance to carry out the required rotation in part (b)(i) using their rectangle from part (a). Part (b)(ii) was often answered poorly, some candidates being confused by the clockwise rotation, when the formula booklet assumes an anticlockwise rotation, and many candidates failing to give numerical values for cos 270° and sin 270°. Most candidates realised that a matrix multiplication was needed in part (c), but many used the wrong matrices or multiplied the matrices in the wrong order.

Q16.

This was a straightforward starter to the paper, yet one that still required a bit of thought; especially with parts (b) and (c), where, respectively, the minus sign and the extra factor of 3 were often overlooked.

Q17.

Rather bizarrely, the final bit of part (a) proved to be the most common difficulty on the paper. Having found the image of $(x, 2x + 1)$ to be $(x - 1, 2 - x)$, hardly anyone managed to deduce that the equation of the image-line was $y = 1 - x$. Parts (b) and (c) were handled very well indeed, although a few candidates tried **BA** rather than **AB** in part (c).

The message as to what is an acceptable way to describe a shear has been stated often in recent years' reports, and this message has clearly been picked up by nearly all centres: the mapping of a point (not on the line of invariant points) to its image was noted far more regularly than in previous sessions, though there are still some rather spurious references to scale factor — 'spurious' in that it is not easy to assign to it an obvious significance or role in the proceedings; indeed, it is not easy even to decide what it is in cases where the shear is not parallel to one of the coordinate axes.

Q18.

High marks were often earned in this question, generally from the multiplication of the matrices rather than from the geometrical explanations, which tended to be shaky. In parts (c) and (d) the vast majority of candidates calculated a matrix product rather than base their answers purely on the transformations already found in parts (a) and (b). The transformation in part (c) was often given as a reflection rather than a rotation, and in part (d) many candidates stated that the matrix was the identity matrix but did not make any geometrical statement as to what this matrix represented. In part (e) the correct matrix was often obtained but the candidates failed to give the correct geometrical interpretation, or in some cases resorted to a full description of the transformation as a combination of the reflection and rotation found in parts (b) and (a). When this was done correctly, full credit was given.

Q19.

The first nine marks in this question seemed to be found very easy for most candidates to earn. The presence of so many k's in the matrices did not prevent them from carrying out all the calculations correctly and confidently, though sometimes the flow of the reasoning in part (b) was not made totally clear. Part (c)(ii), on the other hand, proved too hard for many candidates, who seemed to resort to guesswork for their answers.

Q20.

It was disappointing to see so many candidates unable to find correctly the inverse of a 2 × 2 matrix at the outset of this question. The description in part (b) proved tricky — as had been expected — but actually only relies on understanding of the FP1 work on matrices and transformations. Taking this into account, it was sad to see so few attempts to describe T.

In part (c)(i), many candidates failed to point out the values of p and q that would justify E's status as an ellipse. In most cases, this arose as a result of an inability to divide correctly throughout the given equation by 3. By the time it came to the final two parts, most candidates had made at least one mistake which deprived them of the chance of further successful progress.

Q21.

In this question part (a) was very well answered in general, candidates being able to choose the correct form of matrix for each type of transformation and to insert the correct forms for the necessary sines and cosines.

In part (b), however, the majority of candidates failed to see the close connection between the given matrix and the one that they had found in part (a) (ii). The answers given often appeared to be wild guesses.

In part (c) many candidates saw the need to multiply the two matrices but often in the wrong order. Those who obtained the correct matrix $\begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}$ when giving the geometrical interpretation.

Q22.

Part (a)(i) of this question was almost universally well answered, but part (a)(ii) led to some very slipshod and unclear reasoning. In part (b)(i) many candidates did not appreciate that the scale factor of the enlargement must be the same as the value of q obtained in the previous part. In part (b)(ii) a common mistake was to use tan 60° in finding the gradient of the mirror line instead of halving the angle to obtain tan 30°. Answers to part (c) were mostly very good, even from candidates who had been struggling with the earlier parts of the question.

Q23.

This question was not generally well answered, except for part (b) where most candidates were able to draw the reflected triangle correctly on the insert. In part (a) many candidates seemed not to recognise the transformation as a simple one-way stretch, and even those who did were not always familiar with the corresponding matrix. They could have worked out the matrix by considering the effect of the stretch on the points (1, 0) and (0, 1), but in many cases candidates either gave up, made a wild guess, or embarked on a lengthy algebraic process involving four equations and four unknowns.

In part (c) relatively few candidates saw the benefit of matrix multiplication for the composition of two transformations, and even they often multiplied the matrices the wrong way round. Again it was common to see candidates trying to find the matrix from four equations, but this method was doomed to failure if the candidate, as happened almost every time, paired off the vertices incorrectly.

Q24.

This question was very productive for the majority of candidates, who showed a good grasp of matrices and transformations. A strange error in part (a)(i) was a failure to simplify theexpression $\frac{2\sqrt{3}}{2}$, though on this occasion the error was condoned. The most common mistake in part (a)(ii) was to multiply the two matrices the wrong way round. Only one mark was lost by this as long as the candidate made no other errors and was able to interpret the resulting product matrix as a transformation in part (b)(iii). Occasionally a candidate misinterpreted the 2θ occurring in the formula booklet for reflections, and gave the mirror line as $y = x \tan 60°$ instead of $y = x \tan 15°$.

Q25.

This question was generally well answered. The most common errors were in part (b), where the mirror line was given as $y = x$ rather than $y = -x$. It was, however, acceptable to give this reflection in conjunction with an enlargement with scale factor −3.

In part (c) some candidates omitted the factor 3 before carrying out the matrix multiplication, while others multiplied incorrectly to obtain the 9s and 0s in all the wrong places.

Q26.

Many candidates were successful in part (a)(i) of this question. Wrong answers usually involved a 180° rotation or a reflection in $y = x$. Part (a)(ii) was nearly always done correctly, but the explanations offered in part (a)(iii) were often wrong or too vague to be worth a mark. The matrix calculations in part (b) were done well apart from occasional careless slips. Some candidates seemed to think that three marks were being offered in part (b)(ii) for merely quoting the same answer as in part (b)(i), while others attempted to use matrix algebra to express (**B** + **A**)(**B** – **A**) in an alternative form before evaluating their expression. This usually led to a wrong answer.

Q27.

In part (b), the determinant of the transformation matrix is the scale factor of area enlargement under T – very few candidates used the key word "area" here. In part (d), having found the image of $(x, -x + c)$ to be $(x', y') = (3x - c, 2c - 3x)$, which many successfully did, candidates were supposed simply to show that $y' = -x' + c$. Few candidates understood that this was what was required.

Traditionally, describing transformations is very poorly done, and this was no exception. There was a lack of understanding as to how the algebra, matrix algebra and coordinate geometry all tie together with the geometry of transformations, and some more practice was needed for most candidates on this score.

Q28.

Most candidates carried out the matrix multiplications accurately in part (a) of this question. In part (b) many candidates stated correctly that the required transformation was a rotation, but often gave the angle as 45° anticlockwise rather than clockwise. Attempts at part (c) were rarely completely successful. Many candidates found that $\mathbf{M}^8$ was the identity matrix but then went on to say that $\mathbf{M}^{2006}$ must also be the identity matrix. Others gave the same answer as in part (a) (i), or gave the correct answer without any indication as to their reason for doing so, or with incomplete reasoning.

Q29.

The responses to parts (a) and (b) of this question were most impressive, showing that a good majority of candidates were familiar with matrices of simple transformations. Unfortunately, there were relatively few candidates who knew which way round to multiply the matrices in part (c). Even very strong candidates often went wrong here.

Q30.

Many candidates scored precisely 6 marks out of 10 on this question, finding the coordinates of the image points correctly and drawing an accurate image triangle, but being unable to give either of the answers in part (b). There were even some who earned no marks at all, being apparently unfamiliar with transformations associated with matrices. Those who made a reasonable attempt at part (b) were sometimes working from their diagram and sometimes from the original matrix. The scale factor was very often given as 2 and the angle as 90°, despite the evidence of the diagram.

Q31.

Candidates found this question difficult. The answer to part (a) is the determinant of the given matrix; it was not uncommon for candidates to find this and then take its square root and offer this as the scale factor required. Although there are at least three valid approaches to part (b), most candidates' uncertainty led to a mish-mash of algebra, with the correct answer often embedded in so much other working that the candidate producing it seemed to have no idea that they had just done the necessary work.